

Optimal Portfolio Size Under Parameter Uncertainty

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Nathan Lassance, Rodolphe Vanderveken, Frédéric Vrins

Presented by Kang Guo

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Overview

1. Introduction

2. Theory and Simulation

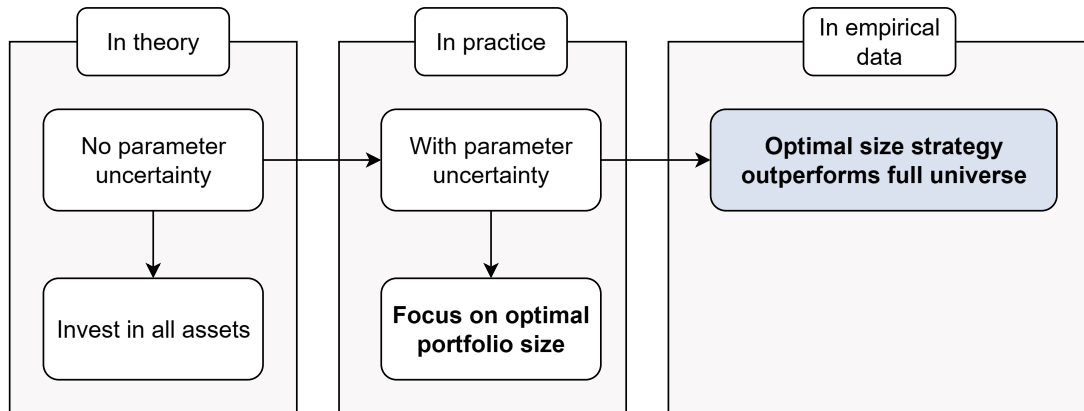
3. Design

4. Results and Replication

5. Ideal

6. Appendix

Summary



Question

- What is the optimal portfolio size to maximize out-of-sample utility under parameter uncertainty?

Background

- Theoretically, investors should invest in all assets
 - Diversify idiosyncratic risk
 - Improve the efficient frontier
- **Challenge:** More assets $\Rightarrow$ More parameters to estimate ($\frac{N^2+3N}{2}$) $\Rightarrow$ Higher estimation risk $\Rightarrow$ Worse out-of-sample performance
- Can optimizing **portfolio size enhance** out-of-sample performance?

Background

- Intuitive solutions: Impose an L_1 penalty on the weights ($\sum |w_i|$)
 - DeMiguel et al., 2009, MS; Yen, 2016, RF; Ao et al., 2019, RFS
- Britten-Jones (1999, JF):

$$\underset{(T \times 1)}{\mathbf{I}} = \underset{(T \times N)}{\mathbf{R}} \underset{(N \times 1)}{\mathbf{w}} + \underset{(T \times 1)}{\mathbf{u}} \quad (1)$$

- $\mathbf{I}$: Vector of ones
- $\mathbf{R}$: Excess return matrix
- $\mathbf{w}$: Optimal weights

Motivation

- These methods suffer from four main drawbacks
 1. **High estimation risk:** All N assets and the penalty parameter on the weights
 2. **Inflexible:** Entangles asset selection and weight optimization into a single step
 3. **Hyperparameter tuning:** Cross-validation adds time and extra estimation risk
 4. **Implicit size:** Portfolio size N is implicit and lacks clear meaning
- **This paper:**
 1. Determine the theoretically optimal portfolio size (N^*)
 2. Optimize portfolio weights exclusively across these N^* assets

Marginal Contributions

- **Theoretically**

- **Prior work:** Assume the number of assets is **exogenously determined** (Markowitz, 1952, JF; Kan and Zhou, 2007, JFQA; Yuan and Zhou, 2024, JFQA).
- **This paper:** Reveal an **inverse U-shaped** relationship between out-of-sample utility and portfolio size.

- **Empirically**

- **Prior work:** Apply constraints to portfolio weights, but still suffer from high estimation risk (Ao et al., 2019, RFS; Yen, 2016, RF; DeMiguel et al., 2009, MS).
- **This paper:** Explicitly fix the optimal size N prior to weight optimization to significantly reduce estimation risk and improve out-of-sample performance.

Optimal Portfolio Size Without Parameter Uncertainty

- Consider an unconstrained MV investor selecting an optimal number of assets $N \leq M$ from a given universe:

$$w^* = \arg \max_{w \in \mathbb{R}^N} U(w) = \arg \max_{w \in \mathbb{R}^N} w' \mu_N - \frac{\gamma}{2} w' \Sigma_N w \quad (2)$$

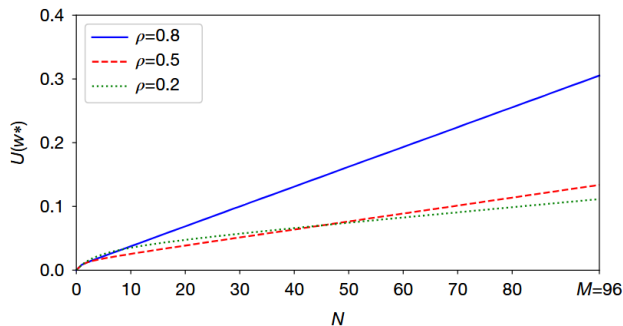
- By solving the first-order condition under an equicorrelation assumption, the **maximum utility can be analytically expressed as a direct function of the portfolio size N :**

$$U(w^*) = \theta_N^2 / (2\gamma), \quad \theta_N^2 = \frac{N}{1 - \rho} \delta_N(\bar{\theta}_N) \quad (3)$$

- Conclusion:** θ_N^2 is the maximum Sharpe ratio. Therefore, **without parameter uncertainty, it is optimal to invest in all available assets ($N = M$)**

Optimal Portfolio Size Without Parameter Uncertainty

- Absent parameter uncertainty, utility increases alongside portfolio size



Optimal Portfolio Size Under Parameter Uncertainty

- Assuming unknown true parameters, an investor relies on a historical sample of size T ($T > N$) with elliptically distributed returns. The expected out-of-sample utility of the sample portfolio $\hat{w}$ is:

$$EU(\hat{w}) = \mathbb{E}[U(\hat{w})] = \mathbb{E} \left[\hat{w}' \mu_N - \frac{\gamma}{2} \hat{w}' \Sigma_N \hat{w} \right] \quad (4)$$

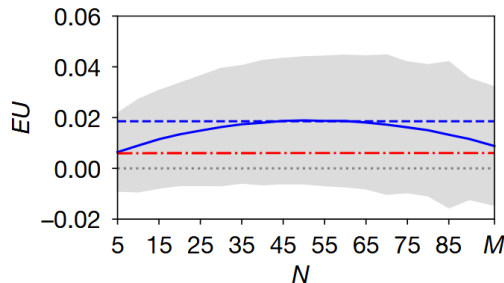
- Assuming stable distributions, expected utility and portfolio size N exhibit an inverse U-shaped relationship:

$$EU(\hat{w}(\alpha^*)) = \frac{1}{2\gamma c_N} \left(\frac{\kappa_{N,1}^2 \theta_N^4}{\kappa_{N,2} \theta_N^2 + \kappa_{N,3} \frac{N}{T}} \right), \quad N^* \approx \frac{T}{2} \quad (5)$$

- The ultimate portfolio performance depends on the trade-off between the increase in **investment opportunities** (as N grows) and the escalating **estimation risk** (as N approaches T)

Simulation Analysis

- **Inverse U-shaped** relationship between portfolio size (N) and out-of-sample utility
- The calculated optimal size N^* is close to the optimal all-asset strategy
 - Solid blue: Utility as a function of N (All-asset strategy)
 - **Dashed blue: Performance at estimated $\hat{N}^*$**
 - Dash-dotted red: $1/N$ benchmark (outperformed)



Data

- Datasets 1–6 (Portfolios): Sourced from Kenneth French and Robert Novy-Marx databases
- Dataset 7 (100STO - Individual Stocks): A random subset of 100 U.S. stocks drawn from a pool of 235

No.	Name	Description	M	Time Period
1	96S-BM	96 portfolios sorted on size and book-to-market	96	July 1963–Aug. 2023
2	108CHA	108 characteristic portfolios built on long and short legs of 54 characteristics in Lassance and Martín-Utrera (2025)	108	Sept. 1966–Dec. 2020
3	100S-OP	100 portfolios sorted on size and operating profitability	100	July 1963–Aug. 2023
4	94IN-NV	48 industry portfolios and 46 characteristic portfolios built on long and short legs of 23 characteristics in Novy-Marx and Velikov (2015)	94	July 1973–Dec. 2013
5	107IN-CHA	47 industry portfolios, 25 portfolios sorted on size and book-to-market, 25 portfolios sorted on operating profitability and investment, and 10 portfolios sorted on momentum	107	July 1963–Dec. 2022
6	98IN-CHA-NV	47 industry portfolios, 25 portfolios sorted on operating profitability and investment, 10 portfolios sorted on momentum, and 16 characteristic portfolios built on long and short legs of eight low-turnover characteristics in Novy-Marx and Velikov (2015)	98	July 1963–Dec. 2013
7	100STO	100 random data sets of 100 U.S. individual stocks	100	Jan. 1998–Mar. 2022

Portfolio Strategies

Name	Description
2F	Two-fund rule that combines the SMV portfolio and the risk-free asset. See equation (8) and Proposition 2 .
3FGMV	Three-fund rule that combines the SMV portfolio, the SGMV portfolio, and the risk-free asset. See equation (A.3) and Proposition A.1 .
3FEW	Three-fund rule that combines the SMV portfolio, the EW portfolio, and the risk-free asset. See equation (A.12) and Proposition A.3 .
EW	Equal-weighted portfolio
EWRF	Combination of EW and the risk-free asset. See equation (OA15) .
SGMV	Sample global minimum-variance portfolio. See equation (A.1) .
GMVRF	Combination of SGMV and the risk-free asset. See equation (24) .
SMV	Sample mean–variance portfolio. See equation (7) .
SMV-ST	SMV portfolio with soft-thresholding and the L_1 -norm-constraint threshold selected via cross-validation. See equation (22) .
SMV-HT	SMV portfolio with hard-thresholding and the weight threshold selected via cross-validation.
F+A	Factor-plus-alpha strategy from Da et al. (2024). See Section OA.4 of the Supplementary Material.

Asset Selection Rules

Name	Description
All	Include all assets: $N = M$.
Rand	Select N assets randomly.
MaxSR	Select N assets with the maximum Sharpe ratios.
MinSR	Select N assets with the minimum Sharpe ratios.
BWSR	Select $\lfloor N/2 \rfloor$ assets with the best Sharpe ratios and $\lfloor N/2 \rfloor$ assets with the worst Sharpe ratios.
MaxVar	Select N assets with the maximum variances.
MinVar	Select N assets with the minimum variances.
BWVar	Select $\lfloor N/2 \rfloor$ assets with the best variances and $\lfloor N/2 \rfloor$ assets with the worst variances.
MinPC	Select the N assets with the minimum correlations with the first principal component.
Best θ_N^2	Select the N assets that deliver the best portfolio Sharpe ratio as in Ao et al. (2019).
MaxW	Select the N assets with the maximum absolute weights.

Out-of-Sample Methodology

- Rebalance portfolios monthly using a 120-month estimation window
- Compare strategies via annualized out-of-sample net utility

$$U_k = 12 \times \left(\hat{\mu}_k - \frac{\gamma}{2} \hat{\sigma}_k^2 \right) \quad (6)$$

- Update weights monthly; refresh optimal N and asset pool annually

Annualized Net Out-of-Sample Utility in Empirical Data

- Optimal- N^* strategies outperform full-universe and naive $1/N$ portfolios

Portfolio	Asset Selection Rule	Data Set						
		96 S-BM	108 CHA	100 S-OP	94 IN-NV	107 IN-CHA	98 IN-CHA-NV	100 STO
2F	All	-0.31	-59.6	-15.3	99.4	-18.8	56.8	-5.47
2F	Rand	12.3	14.2	-7.17	145	1.46	149	-1.77
2F	MaxSR	1.38	11.6	-7.48	23.6	-10.5	-27.1	7.75
2F	MinSR	25.4	73.2	7.10	146	24.9	139	-3.86
2F	BWSR	7.39	84.4	-4.42	133	11.7	242	-29.9
2F	MaxVar	25.0	124	17.9	114	21.7	83.8	-7.25
2F	MinVar	2.47	12.5	-13.1	219	-1.88	225	-0.42
2F	BWVar	13.0	89.3	-6.76	182	0.14	180	-0.53
2F	MinPC	27.4	112	-6.23	115	17.9	94.1	-1.01
2F	Best θ_N^2	-1.22	14.7	-8.77	182	4.73	137	-9.71
2F	MaxW	-127	-305	-179	-581	-129	-366	-77.2
EW	All	8.11	2.8	8.00	3.39	7.13	5.61	13.5

Replication

Portfolio	Asset Selection Rule	Original			Replication		
		96 S-BM	108 CHA	100 S-OP	96 S-BM	108 CHA	100 S-OP
2F	All	-0.31	-59.6	-15.3	-0.43	-41.7	-19.9
2F	Rand	12.3	14.2	-7.17	9.22	17.8	-4.30
2F	MaxSR	1.38	11.6	-7.48	2.21	5.80	-10.5
2F	MinSR	25.4	73.2	7.10	33.0	58.6	10.6
2F	BWSR	7.39	84.4	-4.42	4.43	101.3	-3.09
2F	MaxVar	25.0	124	17.9	35.0	93.0	22.4
2F	MinVar	2.47	12.5	-13.1	3.71	8.13	-17.7
2F	BWVar	13.0	89.3	-6.76	9.10	125.0	-3.38
2F	MinPC	27.4	112	-6.23	32.9	89.6	-9.97
2F	Best θ_N^2	-1.22	14.7	-8.77	-1.83	8.82	-6.58
2F	MaxW	-127	-305	-179	-88.9	-366	-143
EW	All	8.11	2.8	8.00	8.11	2.8	8.00

Extension

- Extend **missing value imputation to the portfolio level**
 - Prior literature predominantly addresses missing values for individual assets, neglecting the portfolio perspective
- Should the T **be an endogenously optimized parameter?**
 - Traditional models treat the historical window (T) as a fixed constant, but it could be endogenized to dynamically balance the bias–variance trade-off

Equivalence: OLS and Mean-Variance Optimization

From Eq.(1), the OLS estimator for the weight vector $\mathbf{w}$ is derived as:

$$\hat{\mathbf{w}}_{OLS} = (\mathbf{R}'\mathbf{R})^{-1}\mathbf{R}'\mathbf{1}$$

Using $\mathbf{R}'\mathbf{R} = T(\hat{\Sigma} + \hat{\mu}\hat{\mu}')$ and $\mathbf{R}'\mathbf{1} = T\hat{\mu}$, the T 's cancel:

$$\hat{\mathbf{w}}_{OLS} = (\hat{\Sigma} + \hat{\mu}\hat{\mu}')^{-1}\hat{\mu}$$

Applying the Sherman–Morrison formula to the inverse:

$$\hat{\mathbf{w}}_{OLS} = \left(\hat{\Sigma}^{-1} - \frac{\hat{\Sigma}^{-1}\hat{\mu}\hat{\mu}'\hat{\Sigma}^{-1}}{1 + \hat{\mu}'\hat{\Sigma}^{-1}\hat{\mu}} \right) \hat{\mu} = \frac{\hat{\Sigma}^{-1}\hat{\mu}}{1 + \hat{\mu}'\hat{\Sigma}^{-1}\hat{\mu}}$$

Normalization: scaling $\hat{\mathbf{w}}_{OLS}$ so that the weights sum to one recovers the sample tangency portfolio:

$$\frac{\hat{\mathbf{w}}_{OLS}}{\mathbf{1}'\hat{\mathbf{w}}_{OLS}} = \frac{\hat{\Sigma}^{-1}\hat{\mu}}{\mathbf{1}'\hat{\Sigma}^{-1}\hat{\mu}}$$

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Two-Fund (2F) Rule

The optimal portfolio weight vector $\hat{\mathbf{w}}_{2F}$ is a linear combination of the risky SMV portfolio and the risk-free asset:

$$\hat{\mathbf{w}}_{2F} = \alpha^* \hat{\mathbf{w}}_{SMV} = \alpha^* \left(\frac{1}{\gamma} \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}} \right)$$

- γ : Risk aversion coefficient.
- $\hat{\boldsymbol{\mu}}, \hat{\Sigma}$: Sample mean vector and sample covariance matrix.

Instead of fully investing in the SMV portfolio ($\alpha = 1$), the rule scales down the exposure based on estimation risk:

$$\alpha^* = c_N \left(\frac{\kappa_{N,1} \theta_N^2}{\kappa_{N,2} \theta_N^2 + \kappa_{N,3} \frac{N}{T}} \right)$$

$EU(\hat{\mathbf{w}}(\alpha^*))$ is Inverse U-Shaped in N

From Eq.(5) , for $N \in \{1, \dots, T-5\}$:

$$EU(N) = \frac{1}{2\gamma} \cdot \frac{\overbrace{\kappa_{N,1}^2 \theta_N^4}^{\text{(A) Investment Opportunities}}}{\underbrace{c_N(\kappa_{N,2} \theta_N^2 + \kappa_{N,3} \frac{N}{T})}_{\text{(B) Estimation Risk}}}.$$

Step 1 — Numerator (A) is at most slowly growing.

θ_N^2 (squared Sharpe ratio) is bounded; $\kappa_{N,k}$ are bounded positive constants. Thus (A) grows at most polynomially.

Step 2 — Denominator (B) diverges as $N \rightarrow T-4$.

With $u = T - N$,

$$c_N = \frac{(T-2)(u-2)}{(u-1)(u-4)} \nearrow +\infty.$$

$EU(\hat{\mathbf{w}}(\alpha^*))$ is Inverse U-Shaped in N

Step 3 — $EU(N)$ is small at both boundaries.

- $N = 1$: No diversification $\rightarrow EU(1)$ is low.
- $N \rightarrow T - 4$: $c_N \rightarrow \infty \rightarrow EU(N) \rightarrow 0$.

Conclusion. $EU(N)$ is strictly positive on the finite discrete domain $\{1, \dots, T - 5\}$, but heavily penalized at both extremes. By the Extreme Value Theorem for discrete sequences, it must attain a *strict interior maximum* N^* :

- $N < N^*$: The growth of signal (A) dominates $\Rightarrow EU \nearrow$
- $N > N^*$: The exploding estimation penalty (B) dominates $\Rightarrow EU \searrow$